CSE 318: Artificial Intelligence Sessional

Online 01

B2

Dual-Mode N-Puzzle with Penalty

Problem

Implement N-Puzzle with two mechanics:

1. **Dual-Mode Tiles**:

   Each numbered tile can move only in one type of direction:

   - **Odd tiles** `1, 3, 5, ...`: only diagonal moves `UL, UR, DL, DR`
   - **Even tiles** `2, 4, 6, ...`: only straight moves `U, D, L, R`

   The blank tile `0` is not odd or even.

   A move is valid only if the numbered tile swapped with the blank is allowed to move in that direction. For example, a straight move involving tile `4` is valid, but a diagonal move involving tile `4` is invalid.

2. **Cumulative Penalty**:

   Each move is either **straight** or **diagonal**.

   - Straight moves: `U, D, L, R`
   - Diagonal moves: `UL, UR, DL, DR`

   The cost of consecutive moves of the same type increases by 1 each time. Whenever the move type changes, the cost resets to 1.

   Example:

   ```
   Moves: U R D
   Types: S S S
   Costs: 1 2 3
   ```

   Example:

   ```
   Moves: U R DL L
   Types: S S D S
   Costs: 1 2 1 1
   ```

Heuristic

For tile t at (r, c) and goal position (r^*, c^*):

$$\Delta r = |r - r^*|, \quad \Delta c = |c - c^*|$$

$$h_t = \begin{cases} \max(\Delta r, \Delta c), & \text{if } t \text{ is odd} \\ \Delta r + \Delta c, & \text{if } t \text{ is even} \end{cases}$$

$$h(n) = \sum_{t \neq 0} h_t$$

The blank tile `0` is excluded. The cumulative penalty is ignored in the heuristic.

Sample

Input:

```
3
1 2 3
4 7 6
8 0 5
```

Output:

```
Minimum cost = 4
Moves: L UR DR

Step 0, Cost = 0
1 2 3
4 7 6
8 0 5

Step 1, Move = L, Move cost = 1, Cost = 1
1 2 3
4 7 6
0 8 5

Step 2, Move = UR, Move cost = 1, Cost = 2
1 2 3
4 0 6
```

```
7 8 5

Step 3, Move = DR, Move cost = 2, Cost = 4
1 2 3
4 5 6
7 8 0
```

Here, `L` is straight, while `UR` and `DR` are consecutive diagonal moves. Their costs are $1, 1, 2$, so the total cost is 4.